MIL-S-29415A(MC)
18 March 1988

SUPERSEDING
MIL-S-29415(MC)
14 June 1984

MILITARY SPECIFICATION

SWEATER; SERVICE WOOL

This specification is approved for use by U. S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for the service wool sweater.

1.2 Classification. The sweater shall be one type and color in the following sizes as specified (see 6.2).

Schedule of Sizes

28/30	38	46
32	40	48
34	42	50
36	44	

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 2.4).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be sue in improving this document should be addressed to Commanding General, Marine Corps Research, Development and Acquisition Command (Attn: Code PSI-G), Washington, D.C. 20380-0001, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC NA . FSC 8405

Distribution Statement A. Approved for public relears; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

V-T-276 - Thread, Cotton
NN-P-71 - Pallet Material Handling, Wood, Stringer Construction, 2 Way and 4 Way (Partial)
DDD-L-20 - Label, for Clothing, Equipage, and Tentage (General Use)
DDD-T-86 - Tape, Textile, Cotton, General Purpose (Unbleached, Bleached or Dyed)

MILITARY

MIL-P-15011 - Pallet, Material Handling, Wood, Post Construction 4 Way Entry
MIL-B-17757 - Boxes, Shipping, Fiberboard (Modular Sizes)
MIL-T-43548 - Thread, Polyester Core: Cotton, Rayon, or Polyester-covered
MIL-C-43665 - Cloth, Wool: Mothproofing Treatment of
MIL-C-43791 - Cloth, Twill, Polyester/Cotton (Durable Press)

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1491 - Glossary of Knitting Imperfections

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitations.

LAWS AND REGULATIONS

U. S. Postal Service Manual

(Application for copies of the manual should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

U. S. DEPARTMENT OF AGRICULTURE

Methods of Test for Grade of Wool

(Application for copies should be addressed to the U. S. Department of Agriculture, Agricultural Marketing Service, Washington, D. C. 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc, Tariff Order Section, 1616 P Street, N. W. Washington, D. C. 20026.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

2.4 Source of documents

2.4.1 Government specifications and standards. Copies of the referenced federal and military specifications and standards are available from the Department of Defense Single Stock Point, Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120. For specific acquisition functions, these documents should be obtained from the contracting activity or as directed by the contracting activity.

3. REQUIREMENTS

3.1 Standard sample. The standard sample is furnished solely for guidance, information and shade to the contractor (see 6.3). The

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

finished sweater shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3). Variation from the specification may appear in the sample, in which case the specification shall govern.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for the first article inspection and approval (see 4.2 and 6.4).

3.3 Materials

3.3.1 Stock. The wool stock shall be fleece or pulled wool, or any combination thereof, not lower than 50's U. S. Standard (see 4.3.1 and 4.3.1.1). The yarn shall be relatively free from short fiber. The use of the reprocessed, reused or waste wool is prohibited.

3.3.1.1 Yarn. The yarn shall be spun on the worsted system from combed top and shall be 2-ply construction.

3.3.2 Knitted fabrics. The knitted component parts of the sweater shall be constructed as specified in 3.3.2.1, 3.3.2.2 and 3.3.2.3 and shall conform to the construction requirements indicated below when tested as specified in 4.4. The same number of ends knitted together shall be in all parts of the garment.

Component Parts	Stitch Type	Wales per Inch (min)	Courses per Inch (min)
Body	2 X 2 rib	6	10
Sleeves	2 X 2 rib	6	10
Cuffs	2 X 2 rib	8	13
Skirt	2 X 2 rib	8	13
Neckband	2 X 2 rib	8	13

3.3.2.1 Body and skirt. The body and skirt (body welt) shall be continuously knitted from 2 X 2 rib construction throughout. The knitting shall be even and regular using .54 inches of yarn per stitch, but the bottom welt rib of the body is to be knitted with .46 inches of yarn per stitch and is to commence with a knitted welt. As a guide only, the above stitch length have been found to give approximately 19 and 24 courses per 2 inches respectively, in a relaxed state. Each body piece is to have between side seams the number of ribs required in Table II. The body and skirt (body welt) shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

knit in one continuous piece to the required body width; when full body width is not possible, one or two side seams shall be permitted. The skirt (body welt) shall finish to the height required in Table II and with a nonraveling bottom edge.

3.3.2.2 Sleeves and cuffs. The sleeves and cuffs shall be of 2 X 2 rib knitted in one continuous piece. The cuff shall finish to the length required in Table II and with a nonraveling bottom edge. The sleeves are to be knitted to the same stiffness as the upper body fabric and the number of ribs in each sleeve near the shoulder are to be as indicated in Table II. The sleeve cuffs are to be knitted as the skirt (body welt) rib.

3.3.2.3 Neckband. The neckband shall be of 2 X 2 rib knitted as tightly as the lower skirt and cuff fabric. The neckband shall conform to the finished width in Table II. The fabric shall be folded lengthwise to form a double neckband 1-1/2 inches wide.

3.3.2.4 Wool content. The finished fabric shall contain not less than 95 percent wool based on the dry weight of the desized specimen when tested as specified in 4.4.

3.3.3 Color. The color of the dyed and finished fabric shall match Olive Green, shade 2247.

3.3.3.1 Matching. The finished knitted fabrics shall match the standard sample under artificial daylight having a color temperature of 7500 degrees Kelvin, and shall be a good approximation to the standard sample under incandescent lamplight at 2800 degrees Kelvin.

3.3.3.2 Colorfastness. The finished knitted fabrics shall show colorfastness to light, dry-cleaning, perspiration, and crocking equal to or better than the standard sample when tested as specified in 4.4. When no standard sample is available, the finished knitted fabrics shall show a minimum of "good" fastness to light, dry-cleaning, and perspiration, and shall show a Munsell Value for crocking not lower than 8.5 when tested as specified in 4.4.

3.3.4 Finish

3.3.4.1 Mothproofing. In the dyeing operation, the wool shall be mothproofed in accordance with MIL-C-43665.

3.3.4.2 Nonfibrous material. The finished knitted fabrics shall contain not more than 3.0 percent chloroform soluble material when tested as specified in 4.4.

3.3.4.3 pH. The pH value of the water extract of the finished knitted fabrics shall be not less than 4.0 nor more than 8.0 when tested as specified in 4.4.

3.3.4.4 Shrink-resistant treatment. The wool shall be treated for resistance to felting shrinkage by a chlorination method approved by

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

the procuring activity. The finished sweater shall not shrink or elongate more than 13.0 percent in length or width when tested as specified in 4.5.1.

3.3.4.5 Alkali solubility. The alkali solubility of the shrink-resistant treated wool shall not be increased by more than 6.0 percent (absolute) over the untreated wool when tested as specified in 4.4.

3.3.5 Tape. The tape for reinforcing the shoulders and lower armholes at side seams shall be cotton, 5/8 inch in width, conforming to type II, class 3 of DDD-T-86, except the minimum filling yarns per inch shall be 58. The tape shall be dyed to approximate the shade of the knitted fabric.

3.3.6 Thread. The thread shall be polyester, cotton, conforming to type IA3, ticket No. 40 or 50, 3 ply, of V-T-276. A cotton covered polyester thread, ticket No. 40 or 50, 2 ply, conforming to MIL-T-43548 may be substituted for the cotton thread. The thread shall be shade Olive Drab T-1, Cable No. 66023.

3.3.6.1 Colorfastness. The dyed thread shall show colorfastness to wet-dry cleaning and perspiration equal to or better than the standard sample when tested as specified in V-T-276 or MIL-T-43548. When no standard sample is available, the dyed thread shall show a minimum of "good" fastness to wet-dry cleaning, and perspiration when tested as specified in V-T-276 or MIL-T-43548.

3.3.7 Label. Each sweater shall have a combination personal, identification, size, and instruction label conforming to type VI, class 10 and 14 of DDD-L-20, except that the inscription shall be shown below and shall show fastness to laundering and dry cleaning:

NOMENCLATURE
NSN
SIZE
NAME OF CONTRACTOR
CONTRACT NO.
DRY CLEAN ONLY
NAME
SERVICE NO.

3.3.8 Cloth for patches. The cloth for the shoulder and elbow patches shall be twill, polyester and cotton, conforming to type II, class 2 or 3 of MIL-C-43791, without durable press treatment. The color shall be Olive Green shade 2248.

3.4 Design. The sweater is a pullover type with a crew neck opening and set-in long sleeves with shoulder and elbow patches (see figures 1 and 2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

3.5 Patterns. The patterns for cutting component parts of the sweater shall be furnished by the contractor, and shall be of proper shape and dimensions to provide good fitting sweaters conforming to the finished measurements specified in Table II.

3.6 Construction. The sweater shall be manufactured in accordance with operation requirements as specified in Table I. The stitch and seam types specified in Table I shall conform to the requirements of FED-STD-751. Where the looper (lower thread) forms a chain effect, the looper shall be on the inside of the sweater. Where a stitch type 500 is specified, the cover thread shall be of 2 ply wool yarn and of the same size as the looper thread. Ends of all seams and stitchings, when not caught in other seams or stitchings, shall be backtacked or overstitched not less than 1/2 inch. Thread breaks (all stitch types) shall be secured by starting the repair stitch not less than 1/2 inch in back of the break. Skipped stitches or thread breaks in 300 or 400 stitch type seams may be repaired by using 301 stitch type.

3.6.1 Shade and size marking. The elbow and shoulder patches for the sweater shall be marked to insure a uniform shade and size on the garment. Any method of marking may be used except:

a. Corrosive metal fastening devices

b. Sew-on tickets

c. Adhesive type tickets whose adhesive mass adheres to or shows discoloration of the material upon removal of the tickets.

TABLE I MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	NDL	BOB/ LPR
1.	Cutting. a. Cut the cloth component parts of the sweater in strict accordance with the contractor's patterns, which shall show shape and directional lines on proper assembly and applicable matching of all parts.					
2.	Replacement of defective components. a. During the spreading, cutting and manufacturing process, cloth component parts having material defects or damages that are classified as					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

TABLE I MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	NDL	BOB/ LPR
2.	Replacement of defective components. (continued)					
	defects in 4.3.2.1 shall be replaced with nondefective and properly matched components.					
3.	Shade and size marking.					
	a. The elbow and shoulder patches shall be marked, ticketed or bundled to insure a uniform shade, size and proper assembly.					
4.	Make neckband. Finished appearance. The neckband shall not be twisted, pulled or distorted in any way. The neckband shall finish 1-1/4 ± 1/8 inches wide.					
	a. Position the ends of the neckband face to face and join with a 1/4-inch seam.	504 or 505	SSh-2 (a)	10-12	A/3	A/3
	b. Fold the neckband in half lengthwise to form a neckband approximately 1-1/2 inches wide and overedge the raw edges.	504 or 505	SSf-3 (a)	10-12	A/3	A/3
5.	Join shoulder seams and attach reinforcement tape.					
	a. Position the front and back parts face to face at the shoulder seams and join together with a 1/4-inch seam.	504 or 505	SSa-2	10-12	A/3	A/3
	b. Attach the reinforcement tape to the shoulder seam from armhole opening to neckline with a 1/4-inch seam.	301 or 401	SSa-a	10-12	A/3	A/3
6.	Attach neckband and set label.					
	a. Position the neckband to neckline and join with a minimum of 1/4-inch seam.	504 or 505	SSh-2 (a)	10-12	A/3	A/3

Source: https://assist.dla.mil - Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

TABLE I MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	NDL	BOB/ LPR
6.	Attach neckband and set label. (continued)					
	NOTE: The joining seam of the neckband shall be positioned 1/2-inch behind the shoulder seam towards the back of the sweater.					
	b. Cover stitch each shoulder seam and the front and back of neckband.	402 or 406	SSh-2 (b)	10-14	A/3	2 ply wool yarn (cover)
	c. Attach label to center back of neck, 1/8-inch below cover seam.	301	SSa-1	10-14	A/3	A/3
7.	Attach shoulder patches. Finished appearance. The shoulder patches shall be attached without any distortion to the shoulder. The shoulder patch shall be centered on the shoulder seam and shall be as wide as the shoulder width of the sweater in a relaxed condition. The shoulder patches shall conform to the finished measurements in Table II.					
	a. The three sides of the patch, centered on the shoulder seam with raw edges turned in a minimum of 1/2 inch, shall be seamed with a double row of stitching 1/16 inch from outer edge and 1/4-inch gage, to the body of the sweater.	301 or 401	SSa-1	10-12	A/3	A/3
	NOTE: The armhole edge of the patch is closed in the sleeve/ arm hole joining seam.					
8.	Make sleeve and attach elbow patch. Finished appearance. The sleeves are to be properly shaped with normal taper between the armhole and cuff. (see Table II).					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

TABLE I MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	NDL	BOB/ LPR
8.	Make sleeve and attach elbow patch. (continued) The knitted rib shall be in a relaxed unstretched condition when attaching the patches. The elbow patch shall be 1/2 to 3/4 inch from sleeve joining seam. a. Position the elbow patch on the backarm of sleeve so that when the patch is joined to the sleeve the patch shall be 1/2 - 3/4 inch from the joining seam and as follows from the bottom edge of cuff: Size 28-42 6 - 6-1/2 inches 44-50 7 - 7-1/2 inches					
	b. Join the elbow patch to the sleeve with a double row of stitching 1/16 inch from outer edge and 1/4-inch gage and with the edges neatly turned under 1/2 inch.	301	LSba-2	10-12	A/3	A/3
	c. Join the sleeve seam with a minimum of 1/4-inch seam, reversing the seam to the outside 3 inches from bottom of cuff. NOTE: Reversing the seam at the lower cuff makes a neater appearance for cuff turn up.	504 or 505	SSh-2 (a)	10-14	A/3	A/3
	d. Cover stitch the sleeve seam for a distance of 3 to 3-1/2 inches from the bottom of the cuff completely covering the outside seam.	402 or 406	SSh-2 (b)	10-14	A/3	A/3
9.	Join sideseams. a. The sideseams shall be joined with a minimum 1/4-inch seam. The edges shall be even at the bottom of the skirt and the armhole.	504 or 505	SSh-2 (a)	10-14	A/3	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

TABLE I MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	NDL	BOB/ LPR
9.	Join sideseams. (continued)					
	b. Turn under stitching chain at bottom of skirt and bartack 3/8 inch from bottom finished edge across the width of seam, catching the chain stitch thread in the bartack.	Bar-tack		21 per bar-tack	A/3	
10.	Attach tape and set sleeves. Finished appearance. The sleeves shall be set without excessive stretching and positioned to the body to lie naturally at an underarm angle to the body of 80-85° when laid on a flat surface. The joining seams are to be neatly and securely finished. The reinforcement tape for the armhole shall be a minimum of 4 inches long.					
	a. Center the 4-inch length of armhole tape on the sideseam and join to the armhole.	301 or 401	SSa-1	10-14	A/3	A/3
	b. With the sleeve seam and sideseam matching, join the sleeve to the armhole, including the shoulder patch raw edge with a minimum 1/4-inch seam.	504 or 505	SSh-2 (a)	12-14	A/3	A/3
11.	Cleaning.					
	a. All thread ends shall be trimmed and loose thread removed throughout the sweater.					
	b. Remove all spots and stains.					
	c. Remove all shade markings.					
12.	Finishing					
	a. The sweater shall be steamed on frames to set the size and remove wrinkles. Care must be taken to insure the measurements stated in Table II are maintained.					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

3.7 Measurements. The sizes and measurements of the finished sweater shall conform to the dimensions specified in Table II. All measurements and tolerances are expressed in inches.

3.8 Weights. The weights of the finished sweater shall be as follows when tested as specified in 4.6. All weights are expressed in pounds.

	Size									
	28/30	32/34	36	38	40	42	44	46	48	50
Wgt/dzn	11	12-3/4	14-1/4	15	17	17-3/4	18-3/4	19-1/2	20-1/4	21

Tolerance -1/4 pounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

TABLE II - FINISHED MEASUREMENTS

Size	A 1/2 Chest	B Back Length (min)	C Skirt Welt Width	D Neck Depth Between Center Front Seam/ Center Back Seam	E Shoulder Point to Shoulder Point	F Neck Width Between Shoulder Seams	G Sleeve Length	H Sleeve width at widest Point of Patch	I Sleeve width at Cuff End of Patch	J Cuff Width	K Cuff Length
28	14	20-1/2	9	1-1/2	15	7	18-1/2	5	4	2-3/4	5-1/2
30	14	20-1/2	9	1-1/2	15	7	18-1/2	5	4	2-3/4	5-1/2
32	14-1/2	21-1/2	9-1/4	2-1/4	15-1/2	7	19	5-1/4	4-1/4	3	5-1/2
34	15	22-1/2	9-1/4	2-1/4	15-1/2	7	20	5-1/4	4-1/4	3	5-1/2
36	15-1/2	23-1/2	9-1/2	2-1/4	16-1/2	7-1/2	20	5-1/2	4-1/2	3	5-1/2
38	16-1/2	24	9-3/4	2-1/4	17	7-1/2	21	5-1/2	4-1/2	3-1/4	5-1/2
40	17	25	10	2-1/4	17-1/2	7-1/2	22	5-3/4	4-3/4	3-1/2	6-1/2
42	17-1/2	26-1/2	10-1/4	2-3/4	18	8	22	5-3/4	4-3/4	3-1/2	6-1/2
44	18-1/2	27	10-1/2	2-3/4	19	8	23	6	5-1/2	4	6-1/2
46	19-1/2	27-1/2	10-3/4	3-1/4	20	8-1/4	23	6	5-1/2	4	6-1/2
48	20	28	11	3-1/4	20-1/2	8-1/4	23	6-1/4	6	4	7
50	20-3/4	28-1/2	11-3/4	3-1/4	21	8-1/4	23-1/2	6-1/4	6	4	7
Tolerance	± 1	± 1	±1/2	± 1/2	±1	±1/2	±1	±1/2	±1/2	±1/2	±1/2

The sweater shall be placed smooth and flat.

A. Measure across front, approximately 1 inch below armhole seams, folded edge to folded edge.
B. The measurement of the back length shall be taken, down the center back, from the bottom edge of the border at the neck to the bottom edge of the sweater.
C. Measure on a straight line across the skirt welt, 1 inch from the bottom, from folded edge to folded edge.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

D. Measure from folded edge of center front collar seamline at bottom of collar to seamline at top of back collar.
E. Measure on a straight line, from shoulder point to shoulder point.
F. Measure across neck opening in line with shoulder seam to shoulder seam at collar.
G. The tolerance for the sleeve lengths shall be consistent. A plus tolerance in one measurement and a minus tolerance in another will not be permitted. The measurement of the sleeve length shall be taken from the armhole seam along the sleeve inseam to the bottom of the sleeve.
H. Measure across sleeve at widest point of elbow patch, from folded edge to folded edge.
I. Measure across sleeve at narrow point of elbow patch, from folded edge to folded edge.
J. Measure on a straight line across sleeve, from folded edge to folded edge.
K. Measure on straight line from bottom of cuff to top of cuff.

TABLE II - Finished Measurements (continued)

Size	No. of Ribs Across Chest (Minimum)	No. of Body Ribs Reduced At Each Armhole	Body Rib Welt Depth (Skirt Height)	No. of Ribs in Sleeve At Armhole Minimum	Armhole Depth	Sleeve Patch Length	Sleeve Patch Width at Widest Point	Sleeve Patch Width At Cuff	Shoulder Patch Length	Shoulder Patch Width
28/30	31	1 - 2	3 - 3-1/2	26	8-1/4	9-1/4	4-1/4	3-1/4	8	AS WIDE
32/34	34	2	3 - 3-1/2	28	8-1/2	9-1/4	4-1/4	3-1/4	8	AS
36	36	2	3 - 3-1/2	31	10	9-1/4	4-1/4	3-1/4	9	PERMITTED
38	38	3	3 - 3-1/2	31	10	9-1/4	4-1/4	3-1/4	9	BY THE
40	40	3 - 4	3-1/2 - 4	34	10-1/2	11	5	3-3/4	9	SHOULDER
42	42	3 - 4	3-1/2 - 4	34	10-1/2	11	5	3-3/4	10	WIDTH OF
44	44	4	3-1/2 - 4	36	11	11	5	3-3/4	10	THE
46	46	4	3-1/2 - 4	36	11	11	5	3-3/4	10	GARMENT
48	48	4	4 - 4-1/2	38	11-1/2	11	5	3-3/4	10	
50	50	4	4 - 4-1/2	38	11-1/2	11	5	3-3/4	10	
Tolerance	±1	±1		±1	±1/2	±1/4	±1/4	±1/4	±1/4	

Source: https://assist.dla.mil -- Downloaded: 2026-1-20T13:00Z
Check the source to verify that this is the current version before use.

3.9 Workmanship. The finished sweater shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels (AQL's).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certification of compliance. Where certificates of compliance are permitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4)
2. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in MIL-STD-1491.

4.4 First article inspections. When first article is required, inspection shall be made of a completely fabricated sweater for all provisions of this specification.

4.5 Quality conformance inspection. Inspection shall be performed in accordance with MIL-STD-105 except where otherwise indicated herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

4.5.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. A certificate of compliance shall be submitted for the wool stock requirements of 3.3.1, yarn requirements of 3.3.1.1, mothproofing requirements of 3.3.4.1, chlorination treatment requirements of 3.3.4.4 and alkali solubility requirements of 3.3.4.5. 1/

4.5.1.1 Wool grade. The wool shall be examined for grade undyed top form by comparison with the applicable U.S. Standard. In the event of a dispute, the wool grade shall be determined by the width method (wedge). The sample unit shall be 1 pound of untreated top. The number of sample units for testing shall be in accordance with Table III. The results from each sample unit shall be reported separately. The lot shall be unacceptable if a single sample unit fails to meet the required grade (see 3.3.1).

Table III- Sampling for Testing

Lot size	Number of sample units
800 or less	2
801 thru 22,000	3
22,001 and over	5

4.5.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine if specified material and manufacturing operations are being adhered to. The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated nonconformance.

4.5.3 Examination of the end item. The end item shall be examined for defects listed in 4.5.3.1 and 4.5.3.2. The applicable inspection levels and AQL's shall be as indicated in 4.5.3.3. The lot size shall be expressed in units of sweaters. The sample unit shall be one finished sweater.

1/ For these characteristics where performance is not by the prime contractor and where a continuous production line involves product of the same character for two or more users (knitters or garment producers), certified test report shall be accompanied by a statement that the (component) lot is entirely furnished from the same production line as the samples(s) tested. Verification test samples shall be drawn from any part of the same production line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

4.5.3.1 General defects. General defects shall be classified as follows:

EXAMINE	DEFECT	CLASSIFICATION		
		MAJOR	MINOR	
			A	B
MATERIAL DEFECTS AND WORKMANSHIP DAMAGES (see Note:)	a. Any hole, cut, tear, mend or, burn	X		
	b. Any drop stitch or run	X		
	c. Heavy yarn or slub more than twice the size of the normal diameter of the yarn		X	
	d. Any knot on outside		X	
	e. Any float		X	
	f. Barre or crock marks		X	
	g. Any pulled or snagged yarn		X	
	h. Burrs or permanent foreign matter	X		
	i. Any permanent distortation (cockled fabric)	X		
	j. Broken yarn	X		
NOTE:	Refer to MIL-STD-1491 for definitions of knitting imperfections			
CLEANNESS	a. Spot or stain on outside (largest dimension in direction)			
	(1) up to 1/4 inch inclusive		X	
	(2) more than 1/4 inch	X		
	b. Thread ends not trimmed or loose			X
	c. One or more shade or size markings not removed			X
	d. Removable foreign matter on outside		X	
COMPONENTS AND ASSEMBLY	a. Any component part or required operation omitted or not as specified (unless otherwise classified herein)	X		
	b. Any component part twisted, distorted, pleated, tight, too full or damaged (unless otherwise classified herein)	X		
	c. Tape not sufficient length for applicable operations	X		
	d. Tape too tight or loose, effecting appearance of sweater	X		

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

EXAMINE	DEFECT	CLASSIFICATION		
		MAJOR	MINOR	
			A	B
	e. Tape not 5/8 inch wide		X	
	f. Tape not positioned properly	X		
	g. Sleeve improperly shaped to provide normal taper. (See Table II for dimensions)	X		
	h. Body improperly shaped to provide normal taper. (See Table II for dimensions)	X		
COLOR OR SHADE	a. Other than specified	X		
	b. Variation in shade between two or more parts	X		
	c. Variations in shade within a part	X		
SEAM(S) AND STITCHING(S)	a. Accuracy of seaming			
	1. Any part of sweater caught in unrelated stitching operation	X		
	2. Seam irregular, twisted or puckered (score only when condition exists on more than 1/2 the length of seam)		X	
	3. Ends of all seams and stitching when not caught in other seams or stitching backtacked or overstitched less than 1/2 inch		X	
	4. Thread breaks not secured or repair stitching starts less that 1/2 inch in back of break		X	
	5. Seam pleated		X	
	6. Line of stitching irregular unevenly gage or not uniform (score only when condition exists on more than 1/2 the length of seam)		X	
	b. Open seams or runoffs (See Notes:)			
	1. Up to 1/4 inch inclusive		X	
	2. More than 1/4 inch	X		

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

EXAMINE	DEFECT	CLASSIFICATION		
		MAJOR	MINOR	
			A	B
NOTE:	One or more broken stitches or two or more consecutive skipped or runoff stitches constitutes an open seam.			
	c. Seam and stitch types			
	1. Not specified seam or stitch type	X		
	2. Any required line of stitching omitted	X		
	3. Any line of stitching not beginning or ending where specified (unless otherwise classified herein)			
	d. Stitch tension	X		
	1. Loose tension resulting in a loose seam	X		
	2. Tight tension (stitches break when normal strain is applied in the direction of the seam or stitching)			
	e. Stitches per inch (score only when condition exists on more than 1/2 the length of seam):	X		
	1. One stitch less than the minimum specified			
	2. Two or more stitches less than the minimum specified	X		
BARTACK(S)	a. Any missing, insecure, misplaced or not serving intended purpose		X	
	b. Length less than 3/8 inch		X	
LABEL	a. Missing, incorrect, or illegible		X	
	b. Misplaced		X	
PATCHES, ELBOW AND SHOULDER	a. Excessive puckering		X	
	b. Patch not proper shape		X	
	c. Patch is not positioned properly	X		
	d. Sleeve elbow patch more than 3/4 inch from sleeve joining seam.	X		

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

4.5.3.2 Dimensional examination. Any sweater measurement deviating from nominal dimensions and tolerance specified in Table II and as follows shall be scored as a measurement defect:

a. Sleeve lengths unequal in length by 1/2 inch or more.
b. Armhole lengths unequal in length by 1/2 inch or more.
c. Skirt welt less than 3 inches or more than 4-1/2 inches in height.
d. Cuff length less than 5-1/2 inches or more than 7-inches in height.
e. Crew neck not within specified tolerance limits.
f. Collar at center back less than 1-1/8 inches or more than 1-5/8 inches in height.

4.5.3.3 Inspection levels and AQL's. The inspection levels and the AQL's expressed in defects per one hundred units (DHU) for visual and dimensional examination, shall be as follows:

	Inspection Level	AQL's Major	Minor A	Total
For visual examination (4.4.3.1)	I	2.5	6.5	10.0
For dimensional examination (4.4.3.2)	S-3	-	-	4.0

4.5.4 Examination of packaging requirements. An examination shall be made to determine that packaging, packing, and marking complies with the Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample units shall be one shipping container fully prepared for delivery except that it need not be closed. Defects of closure listed below shall be examined on shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 DHU's.

Examine	Defect
Marking (Exterior and Interior)	Omitted, incorrect, illegible, improper size, location, sequence or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, improper taping, loose strapping, inadequate stapling, bulging or distortion of container or open and noncontinuous heat sealed seams of polyethylene bag.
Content	Number of sweaters per shipping container is more or less than specified. Size shown on one or more sweaters not same as specified on shipping container. 1/

1/ For this defect, one sweater from each shipping container in the sample shall be examined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

4.6 Testing of the end item. Each shipping lot of the end item shall be tested for the characteristics listed in Table IV in accordance with referenced FED-STD-191 test methods and applicable footnotes. The lot size shall be expressed in units of one sweater each. With the exception of weight per dozen the sample size for testing shall be as follows:

For weight - Three (3) sweaters of each size in the lot regardless of lot size.

All other tests - One (1) sweater

The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements.

Table IV - End Item Tests

Characteristic	Requirement Paragraph	Test Method
Knitted fabrics		
Wales per inch for each part	3.3.2	5070
Courses per inch for each part	3.3.2	5070
Wool content	3.3.2.4	2101
Colorfastness to:		
Light	3.3.3.2	5660
Laundering	3.3.3.2	5614
Dry-Cleaning	3.3.3.2	5621
Crocking	3.3.3.2	5651
Perspiration	3.3.3.2	5680
Nonfibrous content	3.3.4.2	2611
pH	3.3.4.3	2811
Shrinkage	3.3.4.4	7556 1/
Alkali solubility	3.3.4.5	2800 2/
Weight per dozen	3.8	3/

1/ The sweaters shall be initially relaxed in accordance with method 7558.
2/ Three treated and three untreated specimens per sample unit.
3/ The three sweaters of each size shall be allowed to reach equilibrium under standard conditions as defined in FED-STD-191. When under standard conditions, the sweaters shall be weighed as a unit to the nearest 1.0 ounce with the weight obtained projected over a whole dozen. The entire lot of the applicable size shall be rejected if the weight per dozen is below minimum weight specified in 3.8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Each sweater shall be neatly folded to measure approximately 14-1/2 inches in length by 11 inches in width. Each sweater shall then be inserted into a snug fitting clear polyethylene film bag of 0.0015 inch thickness (±20 percent tolerance). The bag shall be formed with heat sealed seams that are straight, continuous, and parallel to each other and the formed edges of the bag. The final closure of the bag shall be heat sealed with the seal made as close as possible to the open end. A 1/4 inch diameter hole shall be made at one corner of the bag to allow excess air to escape. As an alternate, the bag may be fabricated from polyethylene tubing or sheeting and the final closure may be accomplished by means of a tuck or reverse flap.

5.1.2 Commercial. Sweater shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The contractor may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Eighteen sweaters of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled, closed and reinforced conforming to type SF, class weather resistant, grade V2s, size 3A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class weather-resistant, variety DW, grade V15c, of MIL-B-17757. Level A packages shall be packed flat, two in length, one in width, and nine in depth within a shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper.

5.2.2 Level B. Eighteen sweaters of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container assembled and closed conforming to type CF, class domestic, variety SW, grade 275, size 3A of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic, variety DW, grade 200 of MIL-B-17757. Level A packages shall be packed flat, two in length, one in width, and nine in depth within a shipping container. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, W5c, or W6c fiberboard box fabricated in accordance with MIL-B-17757 and closed in accordance with the appendix of the box specification.

5.2.3 Commercial. Sweaters, packaged as specified in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the U.S. Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Polyethylene bagged packages. Polyethylene bagged packages shall have the following information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag so as to permit ready identification.

STOCK
NOMENCLATURE
SIZE
QUALITY
CONTRACT NO
CONTRACTOR

5.4 Palletization. When specified (see 6.2), sweaters packed as specified in 5.2 shall be palletized on a 4-way entry pallet in accordance with load type 1A of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C, K and L or O or P. Pallet pattern shall be in accordance with the appendix of MIL-STD-147. The pallet shall be 4-way, Type I, Class 1, Style 1, Size A, Wood Group I, II, III or IV of MIL-P-15011, or, 4-way, Type IV, V or VIII, Class 1, Style A, Size 2, Wood Group I, II, III or IV, Grade A of NN-P-71. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

6. NOTES

6.1 Intended use. The sweaters covered by this specification are intended for use by personnel of the U. S. Marine Corps.

6.2 Ordering data

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) When first article is not required (see 3.2, 4.2, and 6.4).
(d) Type of approved chlorination method (see 3.3.4.4).
(e) Selection of applicable levels of preservation and packaging and packing (see 5.1 and 5.2).
(f) When optional weather resistant shipping container is required (see 5.2.2.1).
(g) When palletization is required (see 5.4).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data requirements	Applicable DID No.	Options
(a) 4.	Inspection System Program Plan	DI-R-4803	- -
(b) 4.4	First Article Inspection Procedures	DI-T-4901	- -
(c) 4.4	First Article Inspection Report	DI-T-4902	- -
(d) 4.5	Production/Acceptance Inspection Procedures	DI-T-4903	- -
(e) 4.5	Production Inspection Report	DI-T-4904	- -

(Data item description related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

6.3 Standard sample. The standard sample and color sample may be obtained from the procuring activity or as directed by the contracting officer.

6.4 First article. Examination, tests and approval shall be as specified by the contracting officer (see 3.2).

6.5 Subject term (keyword) listing

Service
Sweater
Wool

6.6 Figures. Figures 1 and 2 shows general design of the sweater and are furnished for information only. In the event of conflict between the figures and the text of this specification, the text of this specification shall govern.

6.7 Marginal notations. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing Activity
NAVY-MC
Project No. 8405-N045

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

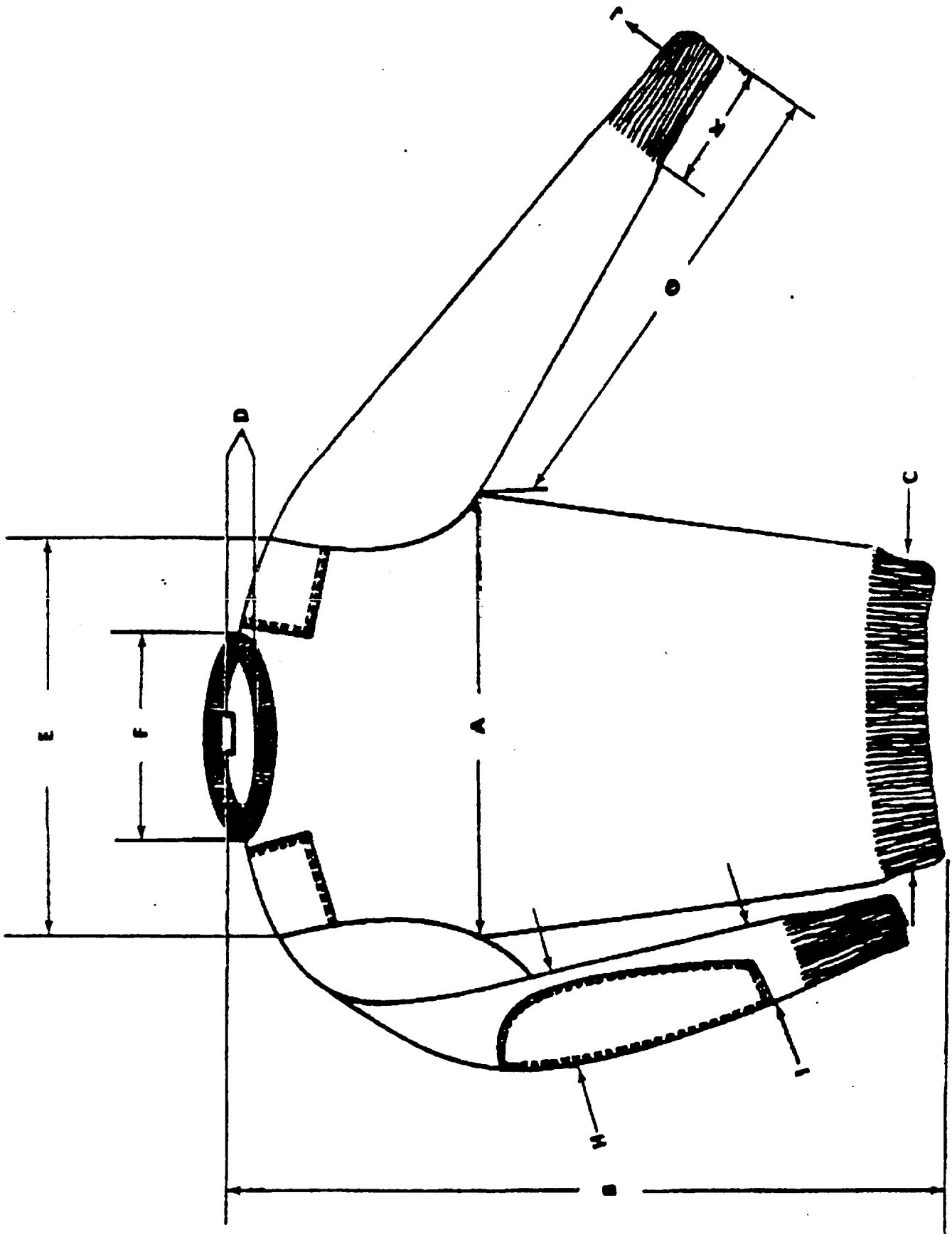


FIGURE 1: SWEATER, SERVICE, WOOL

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

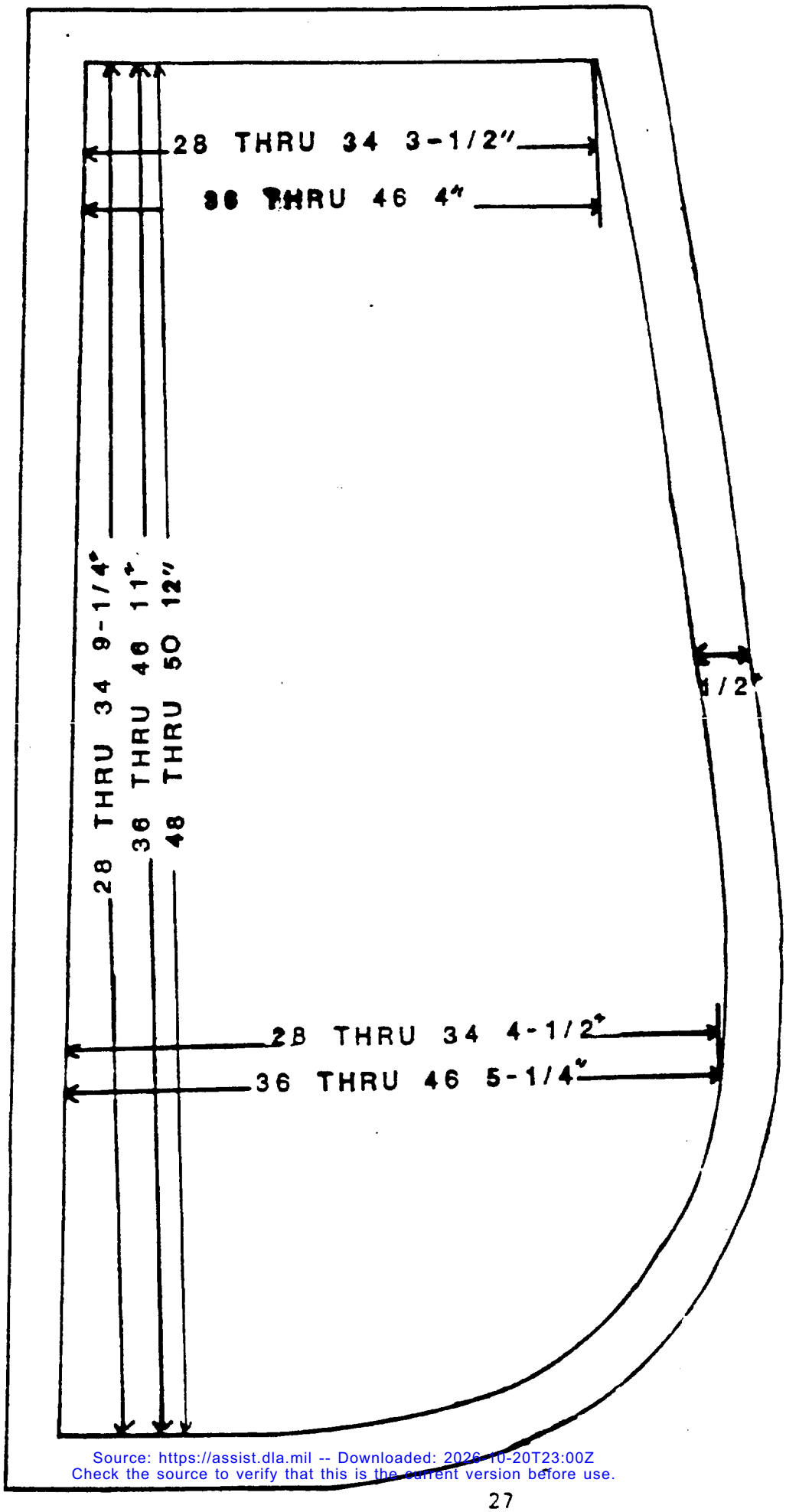


FIGURE 2. SLEEVE PATCH SWEATER, SERVICE WOOL

13050

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding General
Marine Corps Research, Development
and Acquisition Command (PSI-G)
Washington, D.C. 20380-0001

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-29415A(MC)	SWEATER; SERVICE WOOL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:00Z
Check the source to verify that this is the current version before use.